

# Exact WKB framework for quantum resonance and spectral theory

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21/11/2025 Tohoku U Seminar

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- PTEP **2025**, no.10, 103B01 (2025) [arXiv:2503.18741 [hep-th]].
- JHEP **10**, 49 (2025) [arXiv:2505.02301 [hep-th]].
- arXiv:2508.09211 [quant-ph].
- arXiv:2510.11766 [quant-ph].

# Spectral theory

- Finite-dim Hilbert space is well-known
  - ▶ Complete, metric, linear space
- Quantum theory addresses infinite-dim Hilbert space
- ① Discrete spectrum:  $\langle E|E'\rangle = \frac{1}{\mathcal{N}^2}\delta_{E,E'}$
- ② Continuum spectrum:  $\langle E|E'\rangle \propto \delta(E - E')$ 
  - ▶ Let  $\varphi_l$  be regular solution of radial Schrödinger equation
    - ★  $\varphi_l$  is subject to

$$\varphi_l(k, r) \xrightarrow{r \rightarrow 0} j_l(kr) \quad j_l: \text{Spherical Bessel function}$$

with angular momentum  $l$ .

- ▶ At  $r \rightarrow \infty$ , linear combination of Hankel functions  $h_l$ :

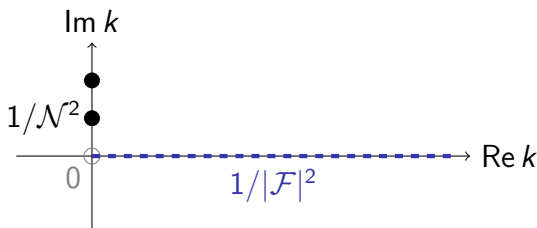
$$\varphi_l(k, r) \rightarrow \mathcal{F}_l(k)h_l^-(kr) - \mathcal{F}_l^*(k)h_l^+(kr) \quad \mathcal{F}_l: \text{Jost function}$$

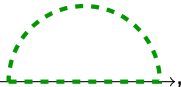
$$\begin{array}{c} \text{blue wavy arrow} \rightarrow h^+ \\ \text{red wavy arrow} \rightarrow h^- \end{array}$$

●  $\longrightarrow r$  Normalize in terms of  $\mathcal{F}, \mathcal{F}^*$

# Spectral function and completeness

- The weight function in inner prod, spectral function, distributes



- From integration of Green function as ,

$$\text{Res: } \underbrace{\sum_i |\varphi_i\rangle\langle\varphi_i|}_{\text{Bound states}} + \underbrace{\int_0^\infty dk |\varphi(k)\rangle\langle\varphi(k)|}_{\text{Continuum states}} = 1$$

- Quantum mechanics works well!

# Backup: Advanced topic: Rigged Hilbert space

- How to treat unbounded operators
  - ▶  $\mathcal{H} \equiv \{\text{square-integrable wavefunctions}\}$
  - ▶ Only on a subdomain  $D(A)$  on which  $A$  is self-adjoint.
- Set of operators  $\{A_i\}$ ; then let us introduce the dense subspace

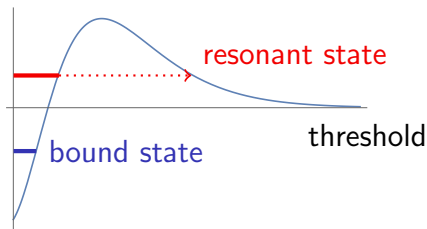
$$\Phi \equiv \cap_i D(A_i) \subseteq \mathcal{H}.$$

- $|x\rangle, |p\rangle$  require larger linear space, so antilinear functional:  $\Phi^\times$ 
  - ▶ For  $|x\rangle \in \Phi^\times$ ,  $\varphi = \int dx |x\rangle \langle x| \varphi \rangle \in \Phi$ .
  - ▶  $\Phi$ : Physical,  $\Phi^\times$ : Functional over  $\Phi$ .
- We find the Gelfand multiplet (rigged Hilbert space)

$$\Phi \subseteq \mathcal{H} \subset \Phi^\times.$$

# Quantum mechanical resonance

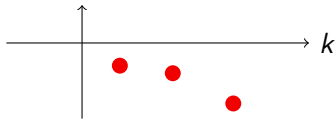
- (Usual) Quasi-stationary state in quantum system
  - ▶ There are some local minima (vacua); one decays to an other.
  - ▶ E.g., double well potential, periodic cosine well, ...
  - ▶ After decaying, finally stable ground state.
  - ▶ Eventually bound state **beyond perturbation theory**.
- Unstable state after decay: **Resonant state**



- ▶ Similar/related to scattering theory, nuclear physics, open system, and non-Hermitian QM

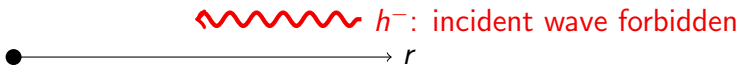
# Problems of resonant states

- Simply say, plane wave in asymptotic region



- But,  $e^{ikr}$  is not in integral contour.
- Non-normalizable wavefunctions:  $e^{ikr} = e^{ik_R r + k_I r} \rightarrow \infty$

- Moreover, at  $r \rightarrow \infty$ , Siegert boundary condition



implies  $\mathcal{F} = 0 \rightarrow$  spectral function? (or ill-def resolvent  $\frac{1}{H-z}$ )

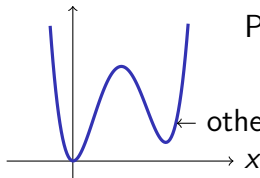
- Completeness?
  - $\sum_{\text{bound}} |\psi_B\rangle\langle\psi_B| + \int_{\text{scattering}} |\psi_S\rangle\langle\psi_S| \stackrel{?}{=} 1$ , but  $|\psi_R\rangle\langle\psi_R|$ ???
- What is the wavefunction itself?
  - Complex probability; expectation value?
  - Transition cross-section?

# Subtleties of resonance physics

- Resonance appears quite universally!
  - ▶ cf. Black hole physics [Miyachi–Namba–Omiya–Oshita '25]
- We aim to understand it in not only math but also phys senses
- There is a large gap between phenomenological success & math
  - ▶ **Complex scaling method**:  $r \rightarrow re^{i\theta}$  for large enough  $\theta$ .
  - ▶ Mathematically well-defined by Aguilar–Balslev–Combes (ABC) theorem [AC '71, BC '71] (as I will explain later).
  - ▶ Physics depends on  $\theta$  explicitly, but computationally successful.
- Quite well-established definition/regularization (historically [ $> 50$ years], experimentally, monumentally)
- To make it transparent, consider the **complex- $x$**  plane
- Focus on analytic solutions of diff eq w.r.t.  $x \in \mathbb{C}$

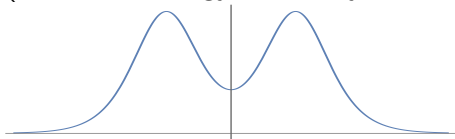
# Resurgence approach to quasi-stationary states

- E.g., for double well potential, perturbation theory suffers from



Perturbative vac decays w/ decay rate  $\Gamma$   
→ Resurgent asymptotic series

- Complex energy of resonance:  $E = E_r - i\frac{\Gamma}{2}$   
(Resonant energy  $E_r$ , decay width  $\Gamma$ )



E.g.,  $V(x) = \frac{U_0}{\cosh^2 \beta(x-a)} + \frac{U_0}{\cosh^2 \beta(x+a)}$

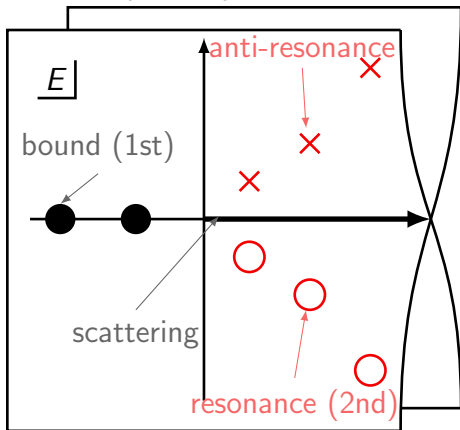
- Can resurgence theory overcome difficulties as **non-perturbative formulation of QM**?
- If so, *essence in “ $\forall$ QM” = analyticity in resurgence*
  - ▶ Quite transparent and precise!!!

# Review I: Resonance as S-matrix pole

- S-matrix is written in terms of Jost function

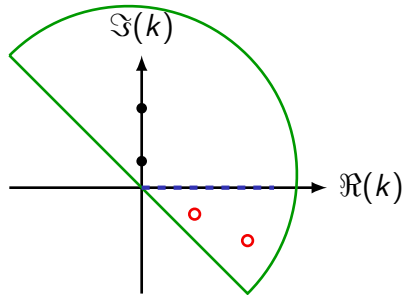
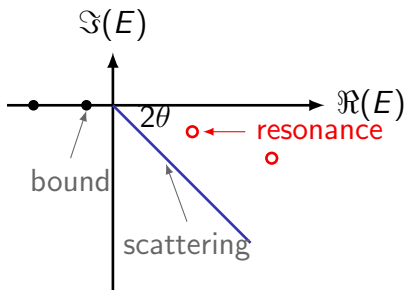
$$S_I(k) = \frac{\mathcal{F}_I^*(k)}{\mathcal{F}_I(k)} \rightarrow \infty \quad \text{pole singularity at resonance}$$

- Riemann surface of complex  $E$ -plane



# Prescription of complex scaling

- Complex scaling method: Radial direction  $r \rightarrow re^{i\theta}$  ( $k \rightarrow ke^{-i\theta}$ )



- ABC theorem [See Myo-Kikuchi-Masui-Katō '14]
  - Resonant wavefunctions are square-integrable.
  - Bound/resonant *energy* is independent from  $\theta$ .
  - Continuous spectrum is rotated by  $2\theta$ .
- Deformation of integral contour of Green func.
  - It includes bound, continuous, and resonant states.
  - $\sum_B |\psi_B^\theta\rangle \langle \tilde{\psi}_B^\theta| + \sum_{R_\theta} |\psi_R^\theta\rangle \langle \tilde{\psi}_R^\theta| + \int_{S_\theta} |\psi_S^\theta\rangle \langle \tilde{\psi}_S^\theta| = 1$  (not \*-ring?)

# Review II: Complexification as non-pert. method

- Rough sketch:

Perturbation theory in quasi-stationary system provides asymptotic series growing factorially,  $\sum_n n! \lambda^n$ .

- 1 Computation with complex coupling  $\lambda \rightarrow e^{\pm i\epsilon} \lambda$ 
  - ▶ Resummation technique  $\rightarrow$  convergent if  $\epsilon > 0$
  - ▶ Ambiguity with  $\epsilon \rightarrow 0 = \pm iO(e^{-1/\lambda})$ : non-perturbative
- 2 Steepest descent around saddle point  $x \in \mathbb{C}$  (Lefschetz thimble)

$$\underbrace{(\text{pert.})}_{\text{finite} \pm \text{ambiguity}} \mp \underbrace{e^{-1/\lambda}}_{\text{nontrivial saddle}} (\text{pert.}) + \dots$$

- Resurgent (no ambiguity)

- Available toolkit here: Exact WKB analysis [Voros]

# More issue: inverted Rosen–Morse potential

- Potential giving resonance

$$V(x) = \frac{U_0}{\cosh^2 \beta(x-a)} + \frac{U_0}{\cosh^2 \beta(x+a)}$$

↓

$$V(x) = U_0 / \cosh^2 \beta x$$

- Exact solution of Schrödinger equation is given by

$$\psi(x) = (1 - \xi^2)^{-\frac{ik}{2\beta}} F\left(-\frac{ik}{\beta} - s, -\frac{ik}{\beta} + s + 1, -\frac{ik}{\beta} + 1, \frac{1 - \xi}{2}\right)$$

$F$ : Gauss hypergeometric function,  $\xi = \tanh \beta x$ ,  $k = \sqrt{2mE}/\hbar$ ,

$$s = \frac{1}{2}(-1 + \sqrt{1 - 8mU_0/\beta^2\hbar^2})$$

- Resonant energy

$$E_n^R = \frac{\hbar^2 \beta^2}{8m} \left[ \sqrt{\frac{8mU_0}{\beta^2 \hbar^2} - 1} - i(2n + 1) \right]^2$$

- **Barrier resonance**: localized state at potential bump (?)
  - ▶ Not intuitively clear why this is happening.

# WKB ansatz

- Consider a Schrödinger equation

$$\left(-\frac{d^2}{dx^2} + \hbar^{-2}Q(x)\right)\psi = 0, \quad Q(x) = 2[V(x) - E]$$

- Introduce WKB ansatz as a formal power series

$$\psi(x, \hbar) = e^{\int^x S(x', \hbar) dx'}, \quad S(x, \hbar) = \frac{1}{\hbar}S_{-1}(x) + S_0(x) + \hbar S_1(x) + \dots,$$

and substitute this into the equation; we have a recursive equation of  $S_i$ : Riccati equation

$$S^2 + \frac{dS}{dx} = \hbar^{-2}Q.$$

- We have two solutions because of the leading-order equation

$$S_{-1}^2 = Q \quad \Rightarrow \quad S_{-1} = S_{-1}^{\pm} \equiv \pm\sqrt{Q}.$$

Solutions are  $\psi^{\pm} = e^{\int S^{\pm}} \sim e^{\pm\frac{1}{\hbar}\int\sqrt{Q}}.$

# Borel resummation

- Borel resummation: summing divergent asymptotic series

$$f(\lambda) \sim \sum_{k=0}^{\infty} f_k \lambda^{k+1} \quad \text{with } f_k \sim a^k k! \text{ as } k \rightarrow \infty$$

⇓ Borel transform

$$B(u) \equiv \sum_{k=0}^{\infty} \frac{f_k}{k!} u^k = \frac{1}{1 - au} \quad (\text{Pole singularity at } u = 1/a).$$

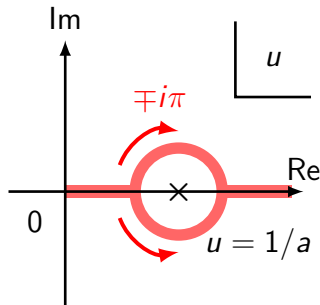
- The Borel sum is given by

$$f(\lambda) \equiv \int_0^{\infty} du B(u) e^{-u/\lambda}.$$

- $a < 0$  (alternating series)  $\rightarrow$  convergent

- $a > 0 \rightarrow$  ill-defined due to the pole

$\Rightarrow$  Imaginary ambiguity  $\sim \pm e^{-1/(a\lambda)}$

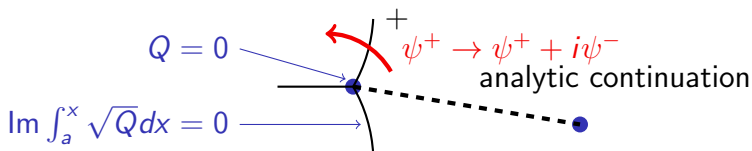


# Stokes phenomena, Stokes geometry

- WKB ansatz is also asymptotic series; where is it Borel summable?
- Integral contour includes  $\mp \int^x dx' \sqrt{Q}$  for  $\psi^\pm$

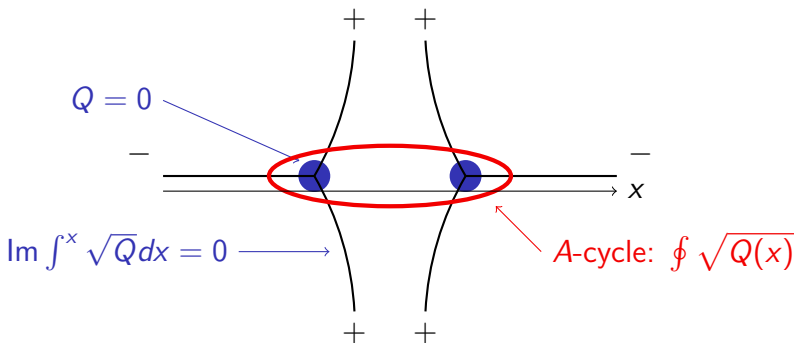
$$\text{Stokes curve : } \left\{ x \left| \operatorname{Im} \int_a^x dx' \sqrt{Q} = 0, \text{ where } Q(x = a) = 0 \right. \right\}$$

- ▶  $a$ : turning point.
- ▶ Not on Stokes curve, Borel summable (analytically continuable).
- ▶ Across a Stokes curve, solution suddenly changes.
- ▶  $\psi^\pm$  is dominant with  $\operatorname{Re} \int_a^x dx' \sqrt{Q} \gtrless 0$ .



# Quantization condition from normalizability

- (Usual) QM: normalizable from  $x \rightarrow -\infty$  to  $x \rightarrow \infty$ 
  - ▶ E.g., Harmonic oscillator

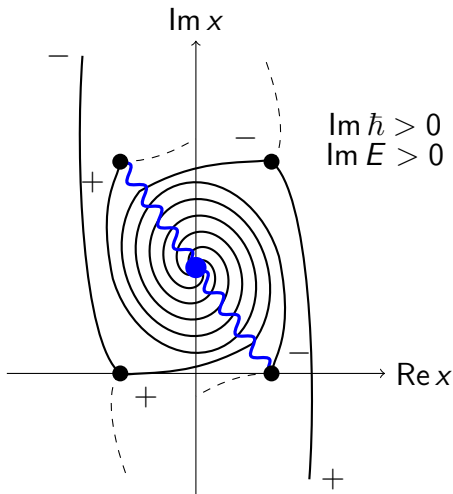
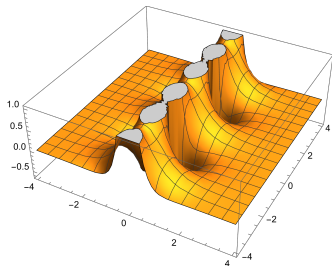
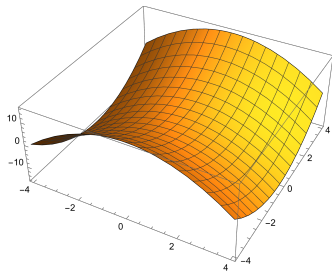


- ▶  $\psi_{\text{final}} = (\text{analytic continuation, Stokes pheno}) \times \psi_{\text{initial}}$
- ▶ Non-normalizable solution vanishes:  $\psi_{\text{initial}}^{\text{damp}} \not\rightarrow \psi_{\text{final}}^{\text{grow}}$

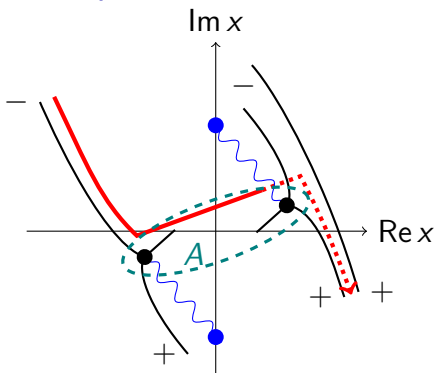
$$1 + A = 1 + e^{\hbar^{-1} \oint \sqrt{Q}} = 0 \quad \Rightarrow \quad E = \hbar \left( n + \frac{1}{2} \right)$$

# How to see resonant state from exact WKB

- Resonant state: quasi-stable in complex region
  - ▶  $V = -x^2$  vs  $V = 1/\cosh^2 x$



# Leading order estimate (Rigorous one in next slide)



- Normalizable path:
- Barrier resonant energy from exact WKB analysis

► A-cycle

$$\int_{-\cosh^{-1} \sqrt{\frac{1}{E}}}^{\cosh^{-1} \sqrt{\frac{1}{E}}} \sqrt{2 \left( \frac{1}{\cosh^2 x} - E \right)} = \sqrt{2}\pi(1 + \sqrt{E})$$

quantization condition:  $1 - A = 0$        $E = \left(1 - \frac{in}{\sqrt{2}}\right)^2$

# Backup: Rigorous estimate = exact solution

- $A = 1$ :

$$\left[ \frac{F\left(-\frac{ik}{\beta} - s, -\frac{ik}{\beta} + s + 1, -\frac{ik}{\beta} + 1, \frac{1-|\xi|}{2}\right)}{F\left(-\frac{ik}{\beta} - s, -\frac{ik}{\beta} + s + 1, -\frac{ik}{\beta} + 1, \frac{1+|\xi|}{2}\right)} \right]^2 = 1,$$

where  $|\xi| = \tanh(\beta \cosh^{-1} \sqrt{U_0/E})$ .

- Here, we use the formula of the hypergeometric function as

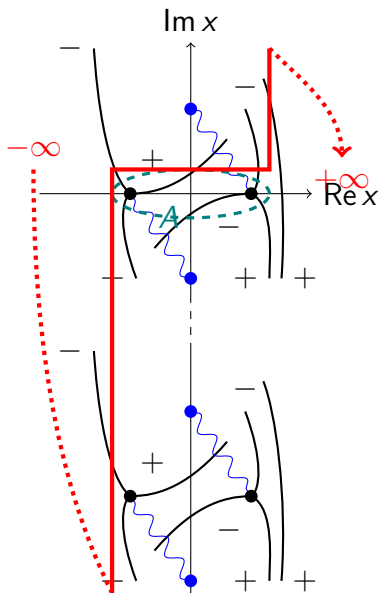
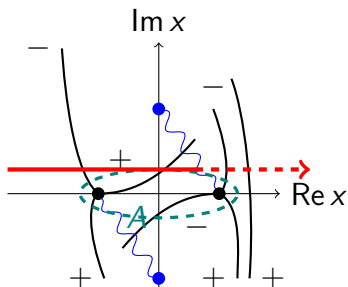
$$\begin{aligned} & F\left(-\frac{ik}{\beta} - s, -\frac{ik}{\beta} + s + 1, -\frac{ik}{\beta} + 1, \frac{1 \pm |\xi|}{2}\right) \\ &= \mathfrak{A} F\left(-\frac{ik}{2\beta} - \frac{s}{2}, -\frac{ik}{2\beta} + \frac{s+1}{2}, \frac{1}{2}, |\xi|^2\right) \mp \mathfrak{B} F\left(-\frac{ik}{2\beta} - \frac{s-1}{2}, -\frac{ik}{2\beta} + \frac{s}{2} + 1, \frac{3}{2}, |\xi|^2\right), \end{aligned}$$

where

$$\mathfrak{A} = \frac{\Gamma\left(-\frac{ik}{\beta} + 1\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(-\frac{ik}{2\beta} - \frac{s-1}{2}\right) \Gamma\left(-\frac{ik}{2\beta} + \frac{s}{2} + 1\right)}, \quad \mathfrak{B} = \frac{\Gamma\left(-\frac{ik}{\beta} + 1\right) \Gamma\left(-\frac{1}{2}\right)}{\Gamma\left(-\frac{ik}{2\beta} - \frac{s}{2}\right) \Gamma\left(-\frac{ik}{2\beta} + \frac{s+1}{2}\right)}.$$

- $\mathfrak{A} = 0, \mathfrak{B} = 0$ : odd/even number of nodes

# What is problematic in naive path? [1/3]



# What is problematic in naive path? [2/3]

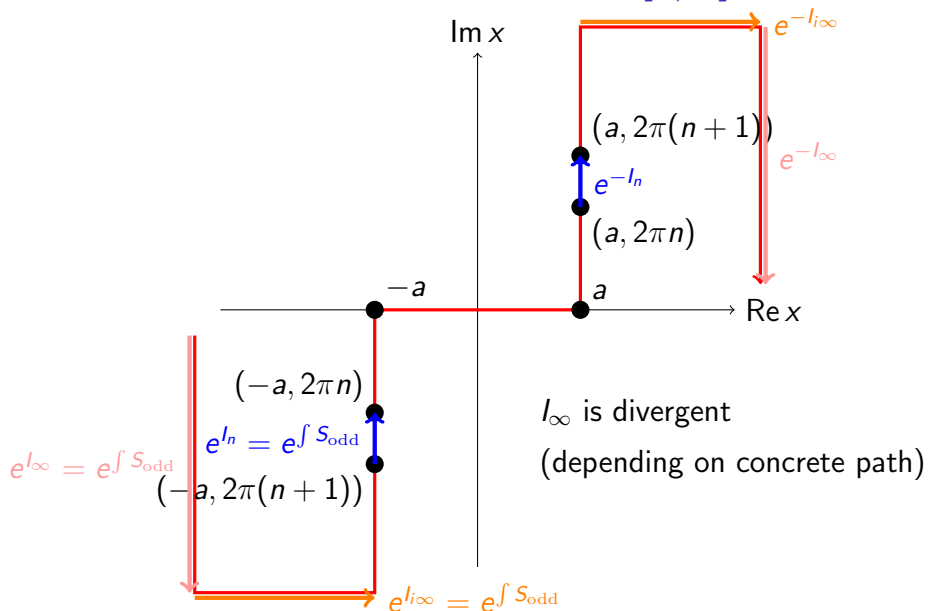
- Total quantization condition is given by

$$\begin{aligned} 0 &= e^{\int_{a+i\infty}^{a+\infty} dx S_{\text{odd}}} \left[ \prod_{n=0}^{\infty} e^{\int_{a+2\pi i n/\beta}^{a+2\pi i(n+1)/\beta} dx S_{\text{odd}}} \right] (1 - A) \\ &\quad \times \left[ \prod_{n=0}^{\infty} e^{\int_{-a-2\pi i(n+1)/\beta}^{-a-2\pi i n/\beta} dx S_{\text{odd}}} \right] e^{\int_{-a-i\infty}^{-a-\infty} dx S_{\text{odd}}} \\ &\stackrel{?}{=} e^{\int_a^{-a} dx S_{\text{odd}}} (1 - A), \end{aligned}$$

►  $S_{\text{odd}} = \frac{1}{\hbar} S_{-1} + \hbar S_1 + \dots$

- Naively, any factor at infinity appears to cancel against each other.

## What is problematic in naive path? [3/3]



# Zel'dovich regularization ['61, Berggren '68]

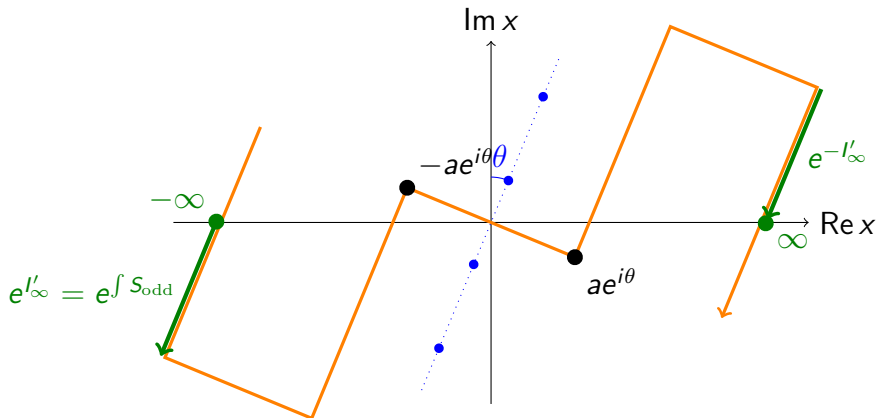
- A simple regularization
- Zel'dovich transformation:  $S = e^{-\epsilon x^2}$

$$\begin{aligned} H\psi = E\psi &\quad \rightarrow \quad H_S\psi_S = E\psi_S, \\ \psi_S &\equiv S\psi, \quad H_S \equiv SHS^{-1}. \end{aligned}$$

- ▶ We find  $H_S = H - \frac{\hbar^2\epsilon}{m} \left( \frac{d}{dx}x + x\frac{d}{dx} \right)$
- Now,  $e^{I_\infty} = e^{\int (S_{\text{odd}} - 2\epsilon x)}$  becomes finite.

# Complex scaling method

- Complex scaling by  $\theta$  rotates the Stokes graph as



- Now,  $e^{I_\infty} \rightarrow e^{I'_\infty}$  becomes finite, and  $A$ -cycle is unchanged.

# ABC theorem

- ① Resonant wavefunctions are square-integrable.
  - ▶ Yes, if the path connects  $\text{Im } x \pm \infty$  or is regularized. It straightforwardly leads to discrete and complex energies.
- ② Bound/resonant energy is independent from  $\theta$ .
  - ▶ Potential and contour are rotated simultaneously;  $A$  is invariant.
- ③ Continuum spectrum is rotated by  $2\theta$ .
  - ▶ To keep Siegert b.c. ( $e^{ikx}$ ),  $k \rightarrow e^{-i\theta} k$  when  $x \rightarrow e^{i\theta} x$ .
  - ▶ Due to  $k = \frac{\sqrt{2mE}}{\hbar}$ ,  $E \rightarrow e^{-2i\theta} E$ .

# Backup: Rigged Hilbert space with resonance

- We introduce

$$\mathcal{D}_\varepsilon^R = \{x \in \mathbb{C} \mid \varepsilon > 0, \lim_{r \rightarrow \pm\infty} |x - r| < \varepsilon\}$$

- ▶  $\mathcal{D}_\varepsilon^R$  is the most crucial singular region.
- if  $x \in \mathbb{C} \setminus \mathcal{D}_\varepsilon^R$ , the Hilbert space is well-defined,  $\mathcal{H}_\varepsilon$ 
  - ▶ Def. of norm and inner product is quite different!
  - ▶ Resonance is included as “bound” state.
- (Operator algebra) Set of operators  $\{A_i\}$  where  $A_i$  is defined on  $D(A_i) \subset \mathcal{H}_\varepsilon$ ; then let us introduce the dense subspace

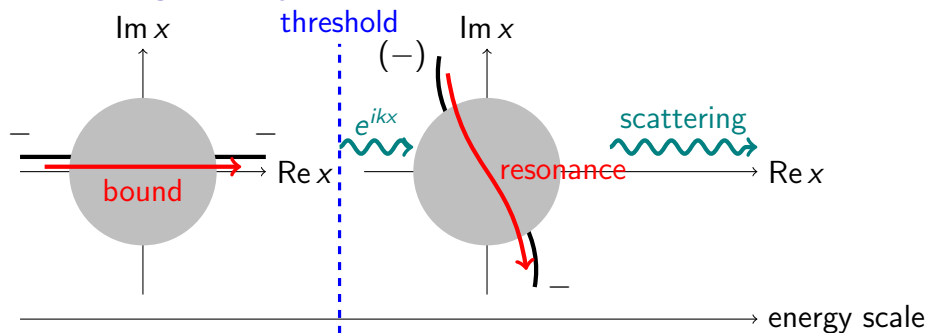
$$\Phi \equiv \cap_i D(A_i) \subseteq \mathcal{H}_\varepsilon.$$

- The range is defined by the limit as  $\varepsilon \rightarrow 0$ , say,  $\Phi^\times$
- We find the Gelfand multiplet

$$\Phi \subseteq \mathcal{H}_\varepsilon \subset \Phi^\times,$$

and  $(\mathcal{H}_\varepsilon, \Phi)$  is the rigged Hilbert space.

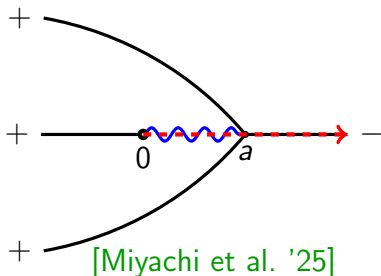
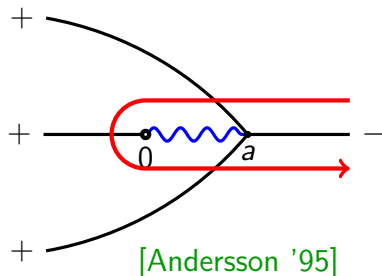
# Scattering theory



- Below threshold: Bound states
- Above threshold: Resonance & continuum spectra
  - ▶ Read transmission/reflection coefficient from monodromy & analytic continuation
  - ▶ Transmission coefficient is divergent if  $E$  comes across resonance.

# Path choice independence in 3D radial equation

- Coulomb Potential  $-\frac{e^2}{r}$ : Closed loop vs. open path



- ▶  $r < 0 \leftrightarrow$  Black Hole: closed path inside event horizon.
- Actually, same phenomena occur!
  - ▶ Aoki–Kawai–Takei work: taking a local (regular) basis near origin, where path starts at  $|r| = \epsilon > 0$  in lateral Borel direction with appropriate small-circle monodromy.
  - ▶ Basis = connected subdominant wavefunction along another lateral Borel direction.

# Summary

- We develop a **unified** framework for analyzing quantum mechanical **resonances** using **exact WKB method**
  - ▶ Serving as a non-perturbative formulation of QM.
  - ▶ Incorporating the Zel'dovich regularization, the complex scaling method, and the rigged Hilbert space.
  - ▶ Demonstrated by examining the inverted Rosen–Morse potential.
  - ▶ Providing construction of physical local basis from closed path.
- Future works:
  - ▶ Realistic potential (E.g.,  $1/\cosh^2(x-a) + 1/\cosh^2(x+a)$ )
  - ▶ Cross-section (Incident wave with connection&monodromy)
  - ▶ Numerical approach
    - ★ Quantization condition  $\rightarrow$  Estimate cycles
  - ▶ Higher dim???